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From the Desk of Dimitri Fane



Dimitri Fane
Director of Product Management, TrakCare

I gave a talk recently **(click here to watch)** where I shared my thoughts on how the speed of innovation in digital health over the past year reminds me of the evolutionary theory of punctuated equilibrium. Punctuated equilibrium holds that, rather than being a slow and steady process, evolution proceeds in bursts, or punctuation events, between long periods of stasis. These punctuation events, we believe, have often been connected to new environmental pressures that jumpstart natural selection. Similarly, the new pressures brought on by COVID-19 have brought about what feels like a punctuation event. The environment in which we all operate has changed radically, and it is organisations, not organisms, that need to adapt in order to survive.

Despite all the havoc and sadness that the pandemic brought, I feel very privileged to have spent the last year watching this punctuation unfold around the world. And like InterSystems, all of our TrakCare clients have adapted to these new pressures and in many cases have thrived. Events continually remind us that the pandemic is far from over and the pain it has caused unfortunately continues, but the innovation and adoption we're seeing gives me a strong sense of optimism for the future.

Along the same lines, TrakCare's pace of change has accelerated. In the last issue of OnTrak, we announced a new naming convention for TrakCare releases in line with our continuous release process. And I'm pleased to say we've now released TrakCare T2021.1, which marks the beginning of this process. You can read more about it in **Issue 7 of OnTrak**, but, in short, this means if you are running TrakCare T2019 or T2020, you are on the current platform and your next patch will increment your version number to T2021.1. And you will receive all the new functionality we have completed each time you patch. We do still expect clients to upgrade the IRIS data platform every couple of years, but once you are on the latest platform, if you stay relatively current there are no more major application upgrades. You'll take on

new functionality incrementally and faster, allowing us to evolve with you and to support you better through this period of punctuation and beyond.

In this issue, you'll find a great summary of new functionality in the past year from our Chief Medical Information Officer, Dr. Hazem El Oraby, and an insightful piece on managed services and cloud hosting from Eugene Rudman, our Director of Managed Services. We also have some great news to share on customer projects. As always, we'd love to hear from you with any questions or feedback and if you have any ideas for topics you'd like to see covered in future newsletters, please do reach out either to me directly or to your local TrakCare team.

Why Healthcare Organisations Should Turn to Managed Services in the Cloud to Support Their Digital Transformation and Improve Efficiencies



Eugene Rudman, Director, Managed Services, InterSystems

The challenges of market volatility, remote workforces and a renewed focus on security, amidst the Covid-19 pandemic, have required organisations worldwide to rethink their business and operational models and become more agile.

Having a reliable IT infrastructure is a crucial part of any organisation's ability to meet these challenges – especially those working in healthcare. For some, so is moving their IT systems to the cloud.

More and more organisations are turning to hosted or managed services to achieve the aims such as those above, and more. But how exactly do they benefit healthcare organisations and support their digital transformation efforts?

Managed application services

Healthcare providers may not have, or may not want to maintain, the expertise required to manage complex, vendor-specific applications. Training and maintaining experts is costly, but without good training and experience, application maintenance, patches and upgrades can all negatively impact patient care.

Having the application vendor manage the application they supply, whether on premises or in the cloud, reduces the need for maintaining such know-how, and instead allows the healthcare organisation to benefit from the vendor's pool of expertise to provide 24x7 monitoring and support.

Ultimately, the application vendor is most likely to have the best knowledge and experience of their application. They can monitor "deeper" inside the application, which enables them to identify and resolve issues better - and before they can impact patient care. They reduce the risk of unwanted outcomes from upgrades and, should anything go wrong, they have unparalleled access to the technical experts, and if needed, right to the software developer to fix the challenging issues. Essentially, problems are resolved quicker, and without the overhead of having to manage an issue across multiple vendors.

By using managed application services, healthcare providers can be more efficient, focus less on managing their application but, instead, spend more time using the application to provide better patient care.

Why have hosted services?

That solves one problem, but another is infrastructure. Healthcare providers increasingly need to exchange health information with others, and patients

increasingly want to be serviced digitally. For several years care providers have wanted, or even needed, to move to the Cloud. In 2017 Market research by the analyst KLAS found that healthcare organisations in the US were increasingly planned to move applications to the Cloud, and in particular, to hosting with their application EPR vendor. This trend proved widespread, and in recent years has been accelerated during and after the current Covid-19 pandemic and high-profile cyber and ransomware attacks. When finally calculated, the cost of the recent cyber-attack suffered by Ireland's Health Service will be considerable.

When facing these demands, healthcare providers with legacy infrastructure find themselves in an unenviable position. Their existing technology may be unreliable and no longer have the features to meet these demands. Cost restraints and a lack of relevant in-house skills prevent them from being able to adequately upgrade it, and they do not have the expertise to secure Internet-facing services, possibly leaving them more vulnerable to security breaches.

A solution is hosted services. This is where a hosting provider provides and manages the infrastructure on which the healthcare organization runs their application. This can be in the public cloud, on a private cloud, or on dedicated servers in the provider's data centres, and the mixing of these deployment methods is called a hybrid cloud.

Hosted services requires the healthcare provider to specify the type and capacity of the resources they require to run their application and they only pay for these resources. The hosting provider ensures the infrastructure remains available 24x7 and manages activities such as hardware maintenance and refresh as required. To reduce the risk of information security breaches the solution provider offers the advanced security controls that are, increasingly, required to protect Internet-facing services.

Other benefits of hosted services includes the opportunities for greater efficiency and scalability, flexible deployment options, access to a wide range of connectivity options, all with the performance normally associated with having dedicated infrastructure resources and expertise.

Why hosted and managed services e.g. "as a service"?

The maximum benefit is obtained from a Cloud deployment when the application vendor provides both the hosted and managed services. In this model, the application vendor provides and manages everything, including the data centres, infrastructure, software, people, processes and tools to deliver the application "as a service" to the healthcare provider.

This model includes benefits of having hosting and managed applications i.e. the healthcare provider no longer has to worry about infrastructure or application maintenance, upgrades, backups, testing High Availability/ Disaster Recovery, access management, and data security. And there are significant benefits gained from the synergies of this unified service delivery model, with cost savings being one key result.

Another key benefit is that the healthcare provider has a single vendor responsible for the end-to-end service. That means one point of contact, supported by a consistent, joined-up set of processes. It prevents having to coordinate across infrastructure and application vendors, or the possibility of things falling through the cracks between the various vendors.

For example, when the vendor makes an infrastructure change, it will be done by someone who understands the impact on the application and will pro-actively take steps to prevent such impact, resulting in higher service availability.

Off-Premises Computing Perception 2017, Energy Shifts to the Cloud, KLAS, December 2017

OnTrak Webinar: "Driving Innovation and Organizational Agility with TrakCare"

During the week of April 26, we highlighted TrakCare innovations that healthcare organisations can leverage to help engage patients and clinicians in the area of telehealth consultations and to augment their "digital front door." If you missed it or want to recap what we discussed, **catch the session on demand here**.

New & Noteworthy in the T2021 Release

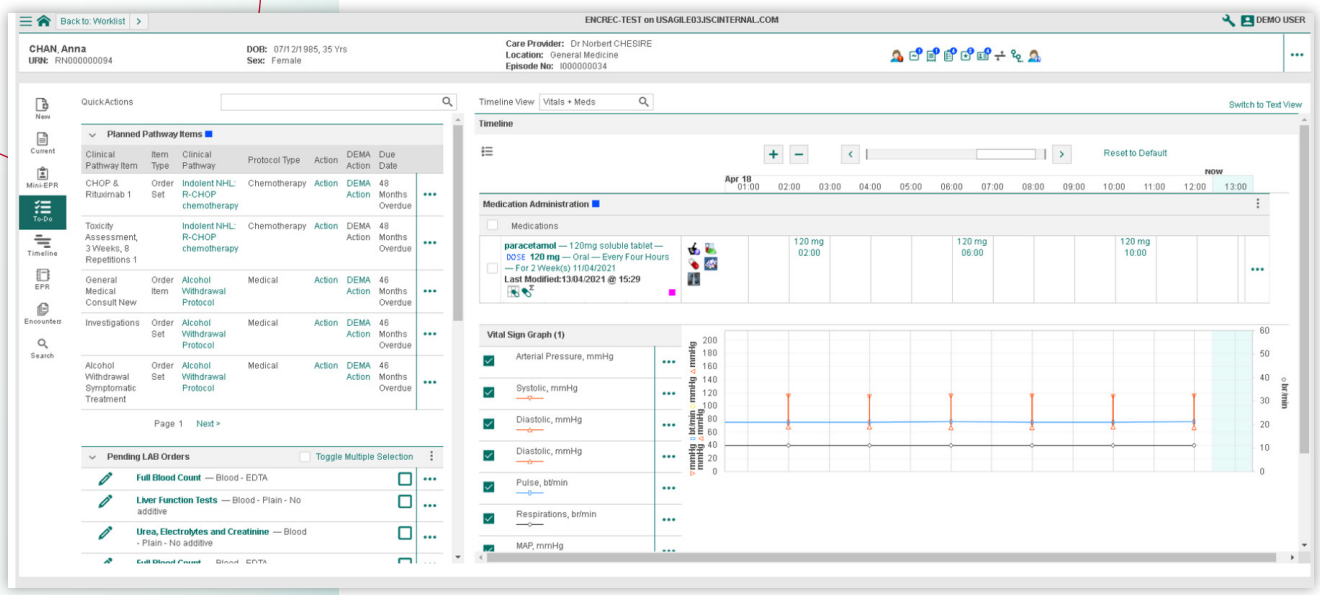
Dr. Hazem El Oraby, Chief Medical Information Officer, TrakCare

Every April, we're excited to release **a new version of TrakCare** with features and enhancements that respond to, and anticipate, customer needs. However, adopting new releases sometimes took longer than we would like which meant waiting to see benefits flow to end users and healthcare organisations.

This year is different. With the recent introduction of our continuous release process, we're now able to make new features available to customers on TrakCare versions from T2019 and beyond, and we'll continue to release new functionality throughout the year as it becomes ready for the upcoming T2022 release.

Here we will highlight some of the most noteworthy T2021 enhancements, many of which will be covered in more depth in future OnTrak articles.

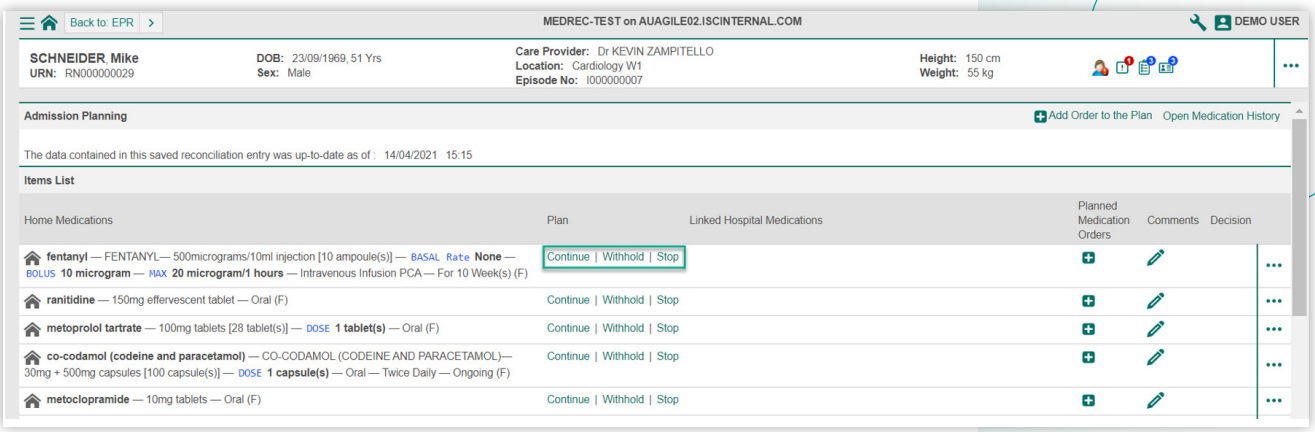
The Encounter Record added significant features and refinements based on our early adopter feedback. You'll immediately notice the integrated clinical timeline that allows users to easily switch between the text view of an Entry and a graphical view while still maintaining access to Entry Actions and mini-EPR. The clinical timeline has seen major improvements and we've also introduced a new 'To Do' list that displays pending orders and clinical pathway tasks, allowing them to be actioned directly from the list. This brings a much tighter integration with the clinical pathways, allowing management of tasks across multiple pathways from a single list. To learn more, see Janet Seaton's story about Encounter Record and Clinical Timeline below.



The Encounter record now has an embedded graphical view (Right) and the new To Do list displaying items from multiple clinical pathways (Left)

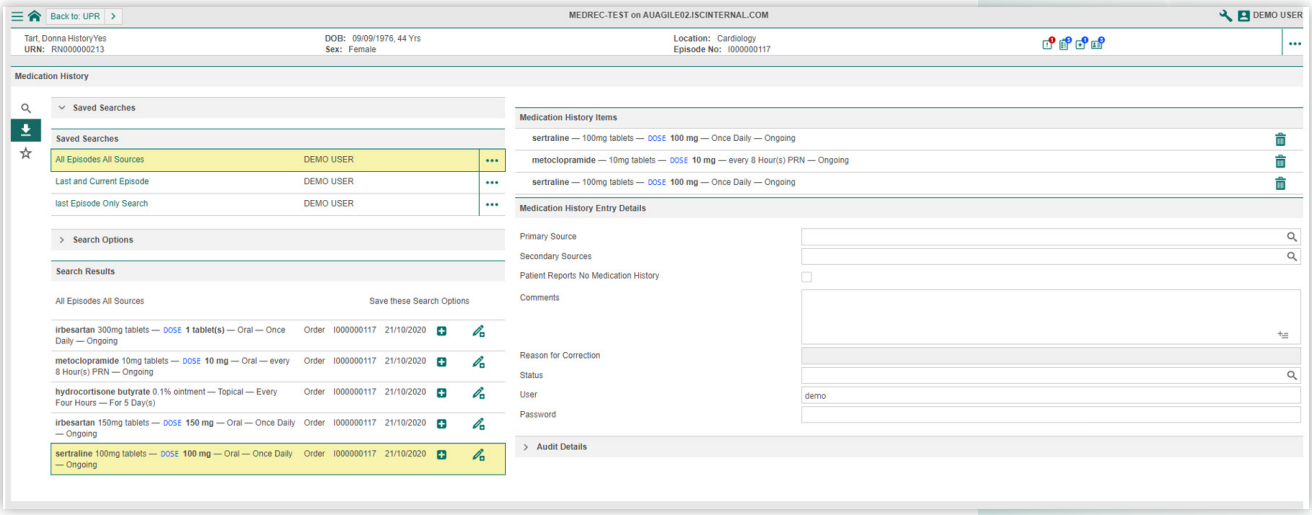
On the usability front, T2021 continues to refine the Mobile Enabled user interface. Finding specific information in a patient record is easier with the ability to filter EPR lists using free text searching across data in all columns, and, being able to group list items (by default or by user selection) is another nifty feature that saves time when trying to understand patient history items in very big records. We are sure clinicians will appreciate the ability to personalise Order Sets so that specific items are preselected the next time the Order Set is needed for a patient.

Medication management also received a big update. The highly anticipated new Medication Reconciliation and Planning functionality provides a means of preventing and rectifying discrepancies between two or more medication lists to ensure a patient's medication list is up to date and accurate. The new, intuitive tool supports recording a reconciliation at Admission and Discharge. Two flavours of workflow are supported: proactive (planning), and retroactive (reconciliation).



The new intuitive medication reconciliation tool supports recording a reconciliation at Admission and Discharge

A new Medication History screen provides multiple ways to view and import medications to the history. Clinicians can also search the full database of medications within TrakCare to record items not available by other means making use of Favourites and Order Sentences, much as they would when placing an Order.



The Medication history screen now has multiple ways of viewing and importing medications into the patient's history

The list goes on with support for Emergency Department Treatment Streams through to Anaesthetic Templates that facilitate accurate and speedy documentation in the Operating Theatre.

We thank you sincerely for your continued feedback and engagement which always helps us to make TrakCare better for all our customers.

Encounter Record & Clinical Timeline Updates

Janet Seaton, Clinical Product Manager, InterSystems

Do you want to see all patient information in one place, not just one system? See details of nurse visits, the last consultation with you (possibly years ago), what your doctor colleague prescribed when your patient attended the emergency department with a broken leg last month, the lab results, the outstanding clinical pathway tasks etc.? And wouldn't it be nice not having to navigate away from the patient record and back again when patient orders, medication administrations or tasks need to be completed? Imagine instead always going back to your open encounter and continuing where you left off.

This is exactly what TrakCare's **Encounter Record** enables. It streamlines all aspects of clinician interaction with TrakCare by providing a single, easy to use workplace for viewing patient-centric records, initiating care activities and plans, and completing necessary clinical documentation. T2021 expands the Encounter Record to provide an even better experience for clinicians. **Mini EPR** provides a convenient patient summary that also makes adding relevant information to new entries easy and transparent. Patients won't miss out on care because upcoming or overdue tasks are clearly identified and can be actioned from the new **To Do List**. Clinicians can quickly toggle between text and timeline views of patient data with the **Clinical Timeline** now available within the Encounter Record (more on Clinical Timeline in a bit). There are also more powerful search options to make finding key information easier than ever.

By providing a common framework for all clinical activity, designed for use by any clinician across all episodes of care and medical specialties, the Encounter Record reduces barriers to adoption and increases speed to proficiency. The result: more time to spend with patients and higher quality medical records to support future health and care initiatives.

Tracking Patient Progress and Planning Upcoming Care is Easier than Ever with Clinical Timeline Embedded in TrakCare's Encounter Record

TrakCare's Clinical Timeline provides synchronised displays of time-orientated data such as patient vital signs, observations, medication administrations, specimen collection and nursing tasks, giving clinicians a focused, time-driven view of patient health conditions and the ability to plan for pending care tasks and interventions. With T2021, Clinical Timeline is embedded in the Encounter Record allowing clinicians to move seamlessly from viewing and documenting narrative or text style records to viewing and documenting information in charts and graphs: all without leaving the Encounter Record. T2021 work focused heavily on getting the structure and navigation right, ensuring all timelines move together, retaining context between displayed items and time periods, and allowing each clinician to see timelines that are appropriate to their specialty. Configuration ensures a timeline relevant to the active encounter type or logged on user is always loaded for easy access and with a relevant time period in focus, while also ensuring users can toggle

between any number of pre-defined timelines (e.g. age/gender-based growth charts) and change the period and interval of the displayed timeline if desired.

Clinical Timeline will allow the concept of flow sheets to be expanded to any care setting supported by TrakCare over time. First among many will be TrakCare Maternity customers who will be able to document labour and delivery entirely from within the Encounter Record beginning from August 2021. As part of that effort, new features such as charting care events, data popovers, and graph zooming will also become available.

Please refer to T2021 release documentation or contact your InterSystems representative to learn more.

Intro to TrakCare 2021

Watch our **Intro to T2021** video for an overview of each new feature and functionality in the T2021 release.

Two More Mediclinic Hospitals in the UAE Implement InterSystems TrakCare for Digital Transformation



Mediclinic Al Jowhara Hospital

Mediclinic Airport Road Hospital in Abu Dhabi and Mediclinic Al Jowhara Hospital in Al Ain have recently implemented InterSystems TrakCare® unified healthcare information system. The implementation at Mediclinic Al Jowhara Hospital marks the 18th go-live of TrakCare across the Mediclinic network of hospitals and clinics in the UAE.

Mediclinic Airport Road Hospital is a **Joint Commission International** accredited facility that offers high standards of primary, secondary and tertiary healthcare services, delivered by a highly qualified team of medical experts. The hospital provides a range of inpatient and outpatient services, including a 24-hour emergency department.

Another key Mediclinic hospital, Al Jowhara Hospital is a premium healthcare facility featuring a full spectrum of medical and surgical specialties. Central to its service offerings is a focus on maternal and child health, providing the latest in obstetric and peadiatric care.

Al Jowhara houses an emergency department, a pharmacy, a laboratory, and a fully equipped radiology unit operating round-the-clock.

"At Mediclinic, we are committed to offering the most comprehensive and evidence-based patient care, delivered in a compassionate manner. Technology innovations empower and enable us to achieve these objectives efficiently to exceed our patients' expectations," said Augustine Amusu, Chief Information Officer at Mediclinic Middle East. "The implementations of TrakCare at Mediclinic Airport Road Hospital and Mediclinic Al Jowhara Hospital bring us closer to our digital transformation goals across the network."

Mediclinic Middle East (MCME), one of the largest private healthcare groups in the UAE, is on schedule to deploy TrakCare across all its hospitals and clinics in the UAE.

TrakCare enables the Group's care providers to have a holistic view of each patient's clinical, administrative, and financial information at any given time. The EMR system supports MCME medical staff in their decision-making, while creating more opportunities to offer patients an enhanced experience and seamless care journey as they spend less time waiting and avoid unnecessary tests.

Advanced interoperability provided by TrakCare enables Mediclinic Airport Road Hospital in Abu Dhabi and Mediclinic Al Jowhara Hospital in Al Ain city to exchange data with Malaffi. This capability also extends to Mediclinic Parkview Hospital which is now able to exchange data with Nabidh.

Malaffi and Nabidh are unified health information exchange platforms that enable healthcare facilities, healthcare professionals and government authorities across the Emirates of Abu Dhabi and Dubai, respectively, to access and share patients' medical information to deliver better healthcare quality and enhance patient safety and overall health outcomes.



Data Powers NHS Glasgow's Fight Against COVID-19

Thanks to early investment in data-informed healthcare, NHS Greater Glasgow and Clyde pulled ahead in the race to respond — and stayed there

Although the pandemic posed challenges for health and care worldwide, NHS Greater Glasgow and Clyde (GGC) in the United Kingdom had prepared for the inevitable. More than a decade ago, the organisation adopted a data-driven and digital-first approach to healthcare delivery that made it easier to reconfigure plans at speed to meet expected volumes, create new treatment pathways, and integrate primary care and other services into a comprehensive response.

Laying the Ground Through Digital Transformation

One of the largest healthcare providers in Europe, NHS GGC serves 1.14 million people and employs 39,000 staff. In 2010, GGC implemented the InterSystems TrakCare® unified healthcare information system for acute services and allowed it to feed the system-wide Electronic Health and Care System (EHCR).

During the pandemic, TrakCare enabled GGC clinicians to manage patients on COVID and non-COVID pathways, providing visibility to essential patient data. What's more, the technology alerted physicians in

near real time when their patients tested positive. Alerts were even visible on TrakCare floor plans, streamlining identification of infected patients.

Using TrakCare, GGC set up six Community Assessment Centres (CAC), where the organisation directed patients experiencing COVID-19 symptoms following triage by phone, and four Specialist Assessment and Treatment Areas (SATA), where GCC conducted further tests and assessments before admitting or discharging them. The organisation created 30 outpatient clinics, 68 ward builds and conversions, and amended intensive-care unit layouts to support the effort. From day one, InterSystems TrakCare enabled GCC staff to connect to all care facilities, CACs, new wards, and even a convention hall turned into a temporary COVID-19 hospital called NHS Luisa Jordan. That connectivity empowered clinicians to coordinate the patient journey in assessment centres, hospitals, and elsewhere.

In the assessment centres, providers used TrakCare to record clinical outcomes. They also placed laboratory requests through TrakCare Order Comms and Workbench functionality.

In the first two months of the pandemic, TrakCare processed more than 1.6 million COVID-19 test results and 16,617 workbench requests. GGC also implemented the system in NHS Louisa Jordan in weeks.

With TrakCare, GGC staff managed workflows and patient treatment pathways from initial diagnosis — including identifying significant comorbidities — to discharge. As teams expanded and patients progressed, new staff used TrakCare to update vital clinical records. And through TrakCare's real-time analytics features, GGC administration leveraged up-to-the-minute dashboards, allowing them to draw critical insights to swiftly respond to the changing situation.

"The way InterSystems has supported us has demonstrated incredible effort, professionalism, and expertise — all of which contributed greatly to helping GGC to respond to the pandemic so rapidly," said Denise Brown, Head of eHealth Strategy & Programmes, eHealth Directorate, NHS Greater Glasgow & Clyde, Scotland.

Next up? GGC plans to leverage TrakCare to expand its telehealth and digital care offerings, in a bid to double down on recent innovations even after the pandemic ends.

After all, GGC wants to respond quickly when the next crisis hits. Just like the healthcare organisation did, in part through its use of TrakCare, during COVID-19.

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